The University of Sydney

Faculties of Arts, Economics, Education, Engineering and Science

MATH2921: Vector Calculus and Differential Equations (Advanced)

Lecturer: Zhou Zhang

Time allowed: 40-Minute In-Person Quiz

This booklet contains **10** pages.

Name (Last, First): Solution ..... SID: .....

Total Marks *(out of 20)*

- Open book. No internet. Individual Work. No discussion.
- Covering material in Weeks 6–9 lectures and practice classes (corresponding to Tutorial Sheets 6–9).
- Time and Venue: Week 11 Practice Class (Thursday, 9 May, 12NOON–1PM), Physics Lecture Theatre (1) 405.
- Style: 5 short answer questions with limited requirements in showing works. Each question with 4 points: 2 final answers for 1 point each and 1 point for relevant works for each.

1. In $\mathbb{R}^3$, consider the vector field $\vec{F} = (x, y, z)$ and the surface

$$S = \{(x, y, z) \mid z = x^2 + y^2,\ \ x^2 + y^2 \leq 1\}.$$

i) Write down the expression of the double integral for the flux of $\vec{F}$ across S in the upward direction: $\boxed{\iint_R (-x^2-y^2)\,dx\,dy}$

ii) The double integral from i) is equal to $\boxed{-\pi/2}$ (numerical answer).

i) $\vec{T}(x,y) = (x, y, x^2+y^2)$.

$\vec{T}_x = (1, 0, 2x)$, $\vec{T}_y = (0, 1, 2y)$

$$\vec{T}_x \times \vec{T}_y = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2x \\ 0 & 1 & 2y \end{vmatrix} = (-2x, -2y, 1)$$ upward

$R = \{(x,y) \mid x^2+y^2 \leq 1\}$

$$\text{flux} = \iint_S \vec{F}\cdot\vec{n}\,dS = \iint_R (x, y, x^2+y^2)\cdot(-2x, -2y, 1)\,dx\,dy$$

$$= \iint_R (-2x^2-2y^2+x^2+y^2)\,dx\,dy$$

$$= \iint_R (-x^2-y^2)\,dx\,dy$$

ii) $$\iint_R (-x^2-y^2)\,dx\,dy = \iint_R -r^2\cdot r\,dr\,d\theta = \int_0^{2\pi} d\theta \cdot (-1)\cdot \int_0^1 r^3\,dr$$

$$= 2\pi \times (-1) \times \frac{1}{4} r^4 \Big|_0^1 = -\frac{\pi}{2}.$$

$R = \{(r,\theta) \text{ polar} \mid \theta \in [0, 2\pi),\ r \in [0,1]\}$

turn to page 3

2. In $\mathbb{R}^3$, consider the vector field $\vec{F}(x,y,z) = (xy, yz, zx)$ and the surface

$$S = \{(x,y,z) \mid x^2+y^2+z^2 = 2, \quad z \geq 1\}.$$

In the following, calculate $\iint_S (\nabla \times \vec{F}) \cdot \vec{n}\, dS$ where $\vec{n}$ is the unit normal vector field of S pointing in the positive z-direction.

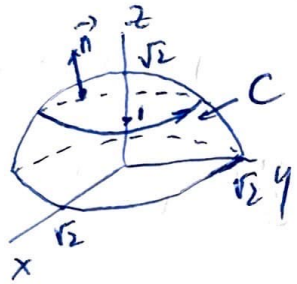
i) Use Stokes' Theorem to transform this surface integral to a line integral

$$\boxed{\int_C \vec{F} \cdot d\vec{r}}, \quad C: \vec{r}(\theta) = (\cos\theta, \sin\theta, 1), \ \theta \in [0, 2\pi)$$

$\vec{F} = (\cos\theta\sin\theta, \sin\theta, \cos\theta)$

making sure to specify the curve including its direction.

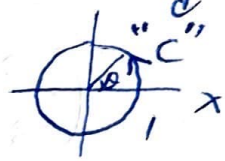
ii) The line integral from i) is equal to $\boxed{0}$ (numerical answer).



i) $\iint_S (\nabla \times \vec{F}) \cdot \vec{n}\, dS$

$= \int_C \vec{F} \cdot d\vec{r}$

projection of C to xy-plane



$z=1$, $x^2+y^2=1$. So C is $\vec{r}(\theta) = (\underset{x(\theta)}{\cos\theta}, \underset{y(\theta)}{\sin\theta}, \underset{z(\theta)}{1})$, $\theta \in [0, 2\pi)$

counter-clockwise when seen from $z > 1$

(rotating from positive x-axis to positive y-axis)

ii) $\int_C \vec{F} \cdot d\vec{r}$

$= \int_0^{2\pi} (\cos\theta\sin\theta, \sin\theta, \cos\theta) \cdot (-\sin\theta, \cos\theta, 0)\, d\theta$

$= \int_0^{2\pi} -\cos\theta\sin^2\theta\, d\theta + \int_0^{2\pi} \sin\theta\cos\theta\, d\theta$

turn to page 4

$= -\int_0^{2\pi} \sin^2\theta\, d\sin\theta + \int_0^{2\pi} \sin\theta\, d\sin\theta \overset{u=\sin\theta}{=} \int_0^0 (\cdots)\, du = 0.$

3. Consider the matrix $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$ and the ODE system

$$\vec{y}\,'(x) = A\vec{y}\,(x).$$

i) Calculate $e^A = \boxed{\frac{1}{2}\begin{pmatrix} e^{-2}+e^{4} & -e^{-2}+e^{4} \\ -e^{-2}+e^{4} & e^{-2}+e^{4} \end{pmatrix}}$ (numerical answer in one matrix).

ii) Calculate the solution of the ODE system with $\vec{y}\,(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

$\boxed{\frac{1}{2}\begin{pmatrix} e^{-2t}+e^{4t} \\ -e^{-2t}+e^{4t} \end{pmatrix}}$ t should be x. ☹

i) $\det(\lambda I - A) = \begin{vmatrix} \lambda-1 & -3 \\ -3 & \lambda-1 \end{vmatrix} = 0.$ $(\lambda-1)^2 - 9 = 0$

$\lambda - 1 = \pm 3$, $\lambda = -2$ or 4.

$\lambda = -2$: $\begin{pmatrix} -3 & -3 \\ -3 & -3 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Pick $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$\lambda = 4$: $\begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Pick $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} -2 & 0 \\ 0 & 4 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}^{-1}$$

$$e^A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} e^{-2} & 0 \\ 0 & e^{4} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} e^{-2} & e^{4} \\ -e^{-2} & e^{4} \end{pmatrix} \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix} e^{-2}+e^{4} & -e^{-2}+e^{4} \\ -e^{-2}+e^{4} & e^{-2}+e^{4} \end{pmatrix}$$

$e^A \rightsquigarrow e^{At}$

$e^{-2} \rightsquigarrow e^{-2t}$

$e^{4} \rightsquigarrow e^{4t}$

turn to page 5

ii) $e^{At} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} e^{-2t} & 0 \\ 0 & e^{4t} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}^{-1} = \frac{1}{2}\begin{pmatrix} e^{-2t}+e^{4t} & -e^{-2t}+e^{4t} \\ -e^{-2t}+e^{4t} & e^{-2t}+e^{4t} \end{pmatrix}$

$\vec{y}(t) = e^{At}\vec{y}(0) = \frac{1}{2}\begin{pmatrix} e^{-2t}+e^{4t} & -e^{-2t}+e^{4t} \\ -e^{-2t}+e^{4t} & e^{-2t}+e^{4t} \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} e^{-2t}+e^{4t} \\ -e^{-2t}+e^{4t} \end{pmatrix}$

4. Consider the 2nd order linear homogeneous ODE

$$x^2y'' + xy' + y = 0.$$

i) Write down the ODE satisfied by the Wronskian $W(x)$ of a fundamental set of solutions for this homogeneous ODE $\boxed{W' = -\frac{1}{x} W}$

ii) For the Wronskian $W(x)$ from i), if $W(1) = 2$, then $W(2) = \boxed{1}$

i) $y'' + \frac{1}{x} y' + \frac{1}{x^2} y = 0$

$p(x) = \frac{1}{x}$.

$W' = -\frac{1}{x} W$.

ii) $W(2) = W(1)\, e^{\int_1^2 -\frac{1}{x} dx}$

$= W(1)\, e^{-\ln x \big|_1^2}$

$= W(1)\, e^{-(\ln 2 - \cancel{\ln 1})}$

$= \frac{1}{2} W(1) = \frac{1}{2} \times 2 = 1.$

turn to page 6

5. Consider the ODE

$$y''(x) - y'(x) = 1.$$

i) Using the method of Undetermined Coefficient or else, find a particular solution for this ODE:

$$\boxed{-x}$$

ii) Find the general solution for this ODE:

$$\boxed{c_1 + c_2 e^x - x}$$

i) $y'' - y' = 0$. $r^2 - r = 0$. $r = 0, 1$.

homogeneous

$e^{0x} = 1$ $\quad e^x$.

Try $y_p = Ax \cdot e^{0x} = Ax$.

Plug into the original ODE:

$(Ax)'' - (Ax)' = 1$

$-A = 1$. $A = -1$. $y_p = -x$.

ii) The general solution is

$y = \underbrace{c_1 + c_2 e^x - x}$ homogeneous general solution

c_1 & c_2 constant

turn to page 7